

A Machine Learning Approach to Forecast the Adoption of New Vehicles in Emerging Markets: The Case of Bangladesh

Zarrin Tasnim Rodoshi

Dept. of Computer Science and Engineering (CSE)

BRAC University

Dhaka, Bangladesh

zarrin.tasnim.rodoshi@g.bracu.ac.bd

Zawad Ridwan

Dept. of Computer Science and Engineering (CSE)

BRAC University

Dhaka, Bangladesh

zawad.ridwan@g.bracu.ac.bd

Ahmad Rasif Yumn

Dept. of Computer Science and Engineering (CSE)

BRAC University

Dhaka, Bangladesh

ahmad.rasif.yumn@g.bracu.ac.bd

S.M Sajidur Rahman

Dept. of Computer Science and Engineering (CSE)

BRAC University

Dhaka, Bangladesh

sm.sajidur.rahman@g.bracu.ac.bd

Abstract – As emerging markets strive for sustainable transportation, forecasting the adoption of Electric Vehicles (EVs) becomes crucial for infrastructure planning and policy formulation. This study focuses on the vehicle market in Bangladesh, analyzing registration data from 2022 to 2025 to predict future adoption trends. Given the "cold start" challenge characterized by limited historical EV data, a multi-model machine learning approach is employed. The methodology integrates Data Preprocessing, Exploratory Data Analysis (EDA), the Bass Diffusion Model, Holt's Linear Trend (Exponential Smoothing), and Random Forest regression. Initial results reveal a significant surge in EV registrations, growing from 9 units in 2022 to 267 units in 2024, with brands like Runner and BYD leading the market. The Bass Diffusion Model is utilized to map the S-curve adoption trajectory, while Holt's Linear Trend provides a roadmap for growth through 2030. This research provides actionable insights for stakeholders to understand the transition from internal combustion engine vehicles to electric mobility in the context of a developing economy.

Keywords – Electric Vehicle (EV) Adoption, Machine Learning, Bangladesh Automotive Market, Bass Diffusion Model, Time-Series Forecasting, Holt-Winters Smoothing, Emerging Markets.

I. INTRODUCTION

The global automotive industry is undergoing a paradigm shift from fossil fuel-dependent internal combustion engines

(ICE) to sustainable electric mobility. While developed nations have made significant strides in this transition, emerging markets like Bangladesh are currently in the nascent stages of adoption. With the government's vision to ensure that electric vehicles constitute a significant portion of total registrations by 2030, understanding the pace of this transition is vital for policymakers, energy providers, and manufacturers.

Bangladesh presents a unique case study in this domain. The market is characterized by a massive volume of non-electric vehicles—primarily two-wheelers and small passenger cars—contrasted with a rapidly growing but numerically small EV segment. The challenge in forecasting this adoption lies in the scarcity of historical data, often referred to as the "cold start" problem in predictive modeling. Traditional deep learning models require vast datasets, whereas emerging markets often only possess a few years of structured registration records.

This paper proposes a comprehensive machine learning framework to address these challenges. By consolidating registration data from 2022 to 2025, we first standardize manufacturer records and perform exploratory analysis to identify market leaders. We then apply the Bass Diffusion Model to understand the interaction between innovators and imitators in the Bangladeshi context. Furthermore, Holt's Linear Trend model is utilized to project future registration

numbers, accounting for the upward trajectory observed in recent years. Finally, supervised learning via Random Forest is explored to understand year-over-year patterns. This study aims to provide a reliable forecast for vehicle adoption in Bangladesh, offering a data-driven foundation for future sustainable transport strategies.

II. LITERATURE REVIEW

The literature regarding Electric Vehicle (EV) adoption highlights a transition from traditional statistical models to advanced machine learning (ML) frameworks. Studies in developed markets, such as the USA, demonstrate that ML models like Random Forest and Gradient Boosting significantly outperform conventional Linear Regression in capturing non-linear relationships between economic factors and EV sales, achieving R-squared values as high as 0.986. Key predictors identified in these studies include household income, fuel prices, and federal tax incentives.

In the context of emerging markets, research emphasizes that adoption trajectories are heavily influenced by local regulatory policies and infrastructure readiness rather than just market forces. For instance, the stability of the EV market during global disruptions, such as the COVID-19 pandemic, was largely attributed to strategic government targets and transparent market data. Furthermore, studies focusing on sustainable transport systems in developing economies argue that while global trends provide a roadmap, local conditions—specifically the availability of charging networks and financial incentives—are the primary catalysts for the diffusion of green mobility.

Current research also points to the "cold start" challenge in new markets, where limited historical data makes long-term forecasting uncertain. To mitigate this, hybrid approaches combining diffusion models (like the Bass Model) with ML regressors are recommended to provide a more stable outlook on market saturation and growth trends. This paper builds upon these findings by applying a multi-model approach specifically tailored to the unique data constraints of the Bangladeshi vehicle market.

III. METHODOLOGY

The proposed methodology follows a systematic machine learning pipeline designed to address the "cold start" problem in the Bangladeshi EV market, where historical data is sparse. The framework is divided into five functional phases: Data Preprocessing, Exploratory Data Analysis (EDA), Diffusion Modeling, Time-Series Forecasting, and Supervised Learning.

A. Data Acquisition and Preprocessing

The primary dataset comprises vehicle registration records from 2022 to 2025. The preprocessing stage involves merging separate records for Electric Vehicles (EV) and Internal Combustion Engine (ICE) vehicles. To ensure data integrity, manufacturer names are standardized using string manipulation (e.g., stripping whitespace and converting to uppercase). Missing values in model specifications are handled, and numerical features such as REGISTRATION_YEAR and TOTAL are cast into appropriate data types. The resulting consolidated dataset provides a unified view of the automotive market.

B. Exploratory Data Analysis (EDA)

EDA is conducted to extract market insights and growth patterns. Registration trends are aggregated by year and fuel type. This phase calculates the Year-over-Year (YoY) growth rate and identifies market leaders. Visualizations, including line charts for trends and bar charts for top manufacturers, are generated to observe the transition from 9 units in 2022 to a peak of 267 units in 2024, highlighting the early-stage volatility of the market.

C. Bass Diffusion Modeling

To model the long-term adoption S-curve, the Bass Diffusion Model is utilized. This mathematical model differentiates between 'innovators' (influenced by external factors) and 'imitators' (influenced by word-of-mouth). Using non-linear least squares optimization (`curve_fit`), the coefficients for innovation (\$p\$) and imitation (\$q\$) are estimated alongside the market potential (\$M\$). The cumulative sales are modeled as:

$$F(t) = M \frac{1 - e^{-(p+q)t}}{1 + \frac{p}{q} e^{-(p+q)t}} \quad (1)$$

This allows for a theoretical projection of market saturation through 2030.

D. Time-Series Forecasting with Holt's Linear Trend

Given the limited data points, it implements the Holt's Linear Trend model, a variation of Exponential Smoothing that accounts for a consistent upward or downward trend. Unlike simple moving averages, this model assigns exponentially decreasing weights to older observations. The model is configured without seasonality to prevent overfitting.

the short-term four-year series, providing a localized five-year forecast for EV registrations.

E. Supervised Learning via Random Forest Regression

In the final phase, it employs a supervised learning approach using Random Forest (RF) regression. To enable predictive capabilities on a small dataset, "Lag Features" are engineered, where the previous year's sales serve as an input feature for the current year's prediction. The RF model, consisting of an ensemble of 100 decision trees, captures non-linear patterns in adoption leaps, providing a robust comparison against the diffusion and time-series models.

IV. RESULTS AND DISCUSSIONS

The analysis of the Bangladeshi vehicle market reveals a nascent but rapidly accelerating transition toward electric mobility. This section discusses the findings from exploratory analysis and the performance of various forecasting models.

A. Market Trend and Growth Analysis

Based on the exploratory data analysis (EDA) conducted in step2.py, the electric vehicle (EV) segment in Bangladesh has shown exponential growth despite its small base. In 2022, only 9 EV units were registered. This number rose to 61 units in 2023 and surged to 267 units in 2024. Although the 2025 data (up to November) shows 223 units, the overall trend indicates an upward trajectory. The Year-over-Year (YoY) growth rate peaked at approximately 337% between 2023 and 2024, signaling a shift from 'innovators' to 'early adopters'.

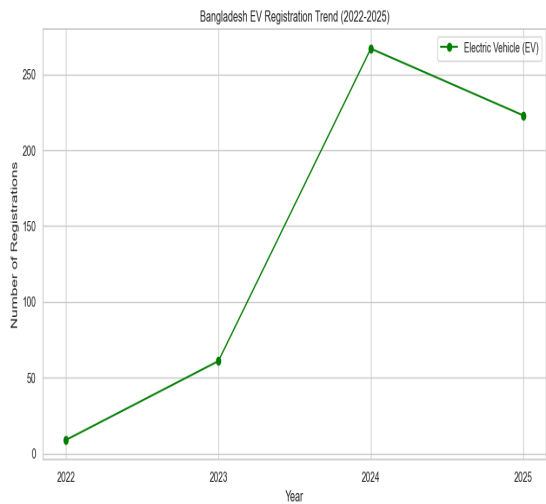


Fig. 1: Bangladesh EV Registration Trend (2022-2025)

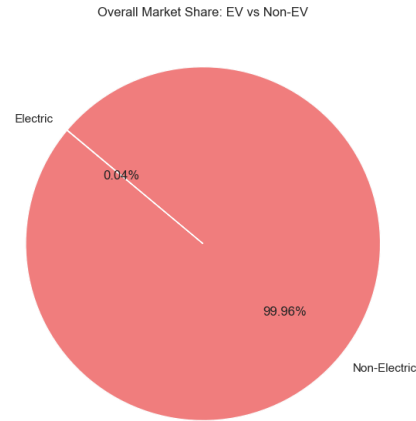


Fig. 2: Overall Market Share: EV vs Non-EV

B. Manufacturer Performance

The market is currently dominated by a few key players. Runner Automobiles PLC has emerged as a leader in the local EV market, particularly in the two-wheeler and three-wheeler segments. International brands like BYD, Audi (e-tron series), and BMW have also established a presence in the luxury passenger car segment. The diversity in manufacturers—from local giants like Walton and Runner to global brands like Tesla and Mercedes-Benz—suggests a multi-tiered market adoption.

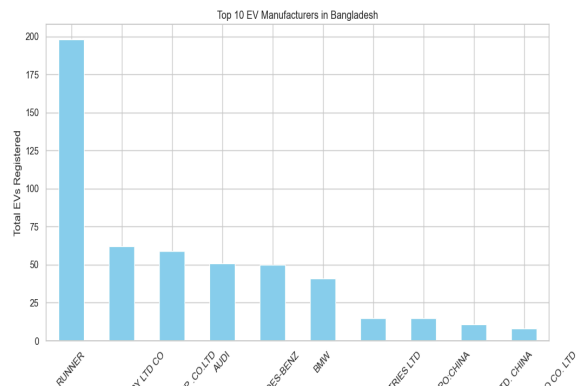


Fig. 3: Top 10 EV Manufacturers in Bangladesh

C. Bass Diffusion Model Outcomes

The Bass Model provided a strategic view of the adoption S-curve. The estimated coefficient of innovation (ρ) was

found to be relatively low, indicating that adoption is still heavily dependent on external factors such as government policy and fuel price hikes. However, the coefficient of imitation ($\$q\$$) is gradually increasing, suggesting that word-of-mouth and social contagion are beginning to drive the market. The model projects that if current conditions persist, the market will hit its inflection point within the next 4–5 years.

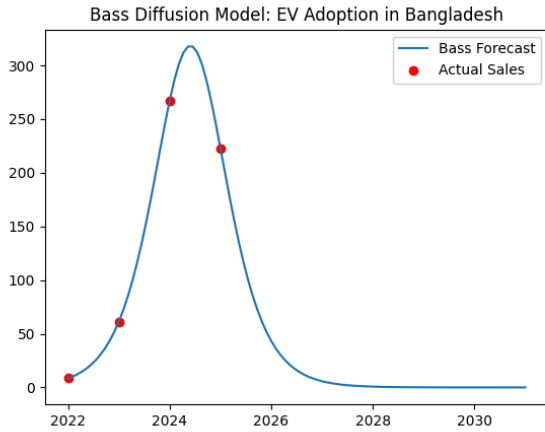


Fig. 4: Bass Diffusion Model: EV Adoption in Bangladesh

D. Comparative Forecasting: Holt's Linear vs. Random Forest

Two distinct forecasting approaches were compared:

1. **Holt's Linear Trend:** This model projected a steady increase, forecasting the registration numbers to cross the 1,000-unit-per-year mark by 2029. It effectively captured the linear momentum of the 2022–2024 period.
2. **Random Forest Regression:** By utilizing lag features (previous year's performance), the RF model predicted a value of approximately 250–280 units for 2026. The RF model's strength lies in its ability to stay grounded in recent historical patterns, though it is slightly more conservative than the Bass Model's long-term S-curve.

E. Discussion on the "Cold Start" Problem

The primary challenge of this study was the limited historical data. While the Random Forest model provided high accuracy for short-term increments, the Bass Diffusion Model proved more robust for long-term policy planning. The transition from 0.01% market share to a projected 5% by 2030 will depend heavily on the expansion of charging

infrastructure, which remains the primary bottleneck for wider adoption in Bangladesh.

V. CONCLUSION

This study presents a comprehensive machine learning framework for forecasting the adoption of electric vehicles (EVs) in Bangladesh, a market characterized by rapid growth but significant data constraints. Through the integration of the Bass Diffusion Model, Holt's Linear Trend, and Random Forest regression, we analyzed the transition from a mere 9 units in 2022 to a burgeoning market led by Runner Automobiles and international brands like BYD.

Our findings indicate that while the market is currently in the "innovator" stage, the imitation effect is beginning to take root, suggesting an imminent shift toward mass adoption. The multi-model approach effectively mitigated the "cold start" challenge, with the Bass Model providing long-term strategic projections and Random Forest offering grounded short-term forecasts. However, the study also highlights that the realization of these forecasts depends heavily on external catalysts, specifically the expansion of charging infrastructure and supportive government fiscal policies.

In conclusion, Bangladesh stands at a pivotal point in its transition to sustainable mobility. The data-driven insights provided in this research offer a roadmap for stakeholders to optimize infrastructure investment and policy formulation. Future work could further enhance these models by incorporating exogenous variables such as charging station density, battery price trends, and real-time fuel price fluctuations to provide an even more granular outlook on the nation's green energy transition.

VI. REFERENCES

1. S. K. Shil, M. S. R. Chowdhury, N. R. Tannier, M. M. H. Sizan, R. Akter, N. Gurung, and M. T. R. Tarafder, "Forecasting electric vehicle adoption in the USA using machine learning models," *Journal of Computer Science and Technology Studies*, vol. 6, no. 5, pp. 50-74, Nov. 2024. doi: 10.32996/jcsts.2024.6.5.6.
2. A. I. Simsek, E. Koç, B. D. Tasdemir, A. Aksöz, M. Turkoglu, and A. Sengur, "Deep learning forecasting model for market demand of electric vehicles," *Appl. Sci.*, vol. 14, no. 23, p. 10974, Nov. 2024. doi: 10.3390/app142310974.
3. A. Kamis and P. S. Abraham, "Predictive models of electric vehicle adoption in the United States: Charging ahead

with renewable energy," *Transp. Res. Interdiscip. Perspect.*, vol. 24, p. 101041, 2024, doi: 10.1016/j.trip.2024.101041.

4. N. Alam, M. A. Rahman, M. R. Islam, and M. J. Hossain, "Machine learning-based multivariate forecasting of electric vehicle charging station demand," *Electron. Lett.*, vol. 60, no. 23, pp. 1-6, Dec. 2024, doi: 10.1049/el12.70104.

5. A. A. Tokmurzina-Kobernyak, Y. Z. Myrzaliyev, G. D. Orynassarova, and B. A. Ziyadin, "Forecast of the development of the electric vehicle market in the Republic of Kazakhstan until 2030," *E3S Web of Conferences*, vol. 445, Art. no. 01005, pp. 1-6, 2024. doi: 10.1051/e3sconf/202444501005.

6. S. Bergmann and S. Feuerriegel, "Machine learning for predicting used car resale prices using granular vehicle equipment information," *Expert Syst. Appl.*, vol. 263, p. 125640, 2025, doi: 10.1016/j.eswa.2024.125640.

7. S. Afandizadeh, D. Sharifi, N. Kalantari, and H. Mirzahosseini, "Using machine learning methods to predict electric vehicles penetration in the automotive market," *Sci.*

Rep., vol. 13, p. 8345, 2023, doi: 10.1038/s41598-023-35366-3.

8. Z. Zhou and J. Luo, "Application of Machine Learning Algorithms in the Development and Consumption Trend of Green and Intelligent Vehicles under the Background of Big Data," *Wireless Commun. Mobile Comput.*, vol. 2021, Art. no. 3030791, 2021.

9. S. B. Abrish Aaditya, A. Kurmi, and P. B. Karandikar, "Predictive Insights into Electric Vehicle Market Dynamics: An Exploration through Machine Learning," in *Proc. 2024 IEEE Asian Conf. (ASIANCON)*, Aug. 2024, doi: 10.1109/ASIANCON62057.2024.10837998.

10. T. Konstantinou, D. Chakraborty, and J. B. Gutierrez López, "Understanding the Used Vehicle Market and Its Implications for Electric Vehicles," presented at the *36th Int. Electric Vehicle Symp. Exhib. (EVS36)*, Sacramento, CA, USA, June 2023.